Symbiosis Institute of Technology, Nagpur

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Established under Section 3 of the UGC Act, 1956 vide notification No. F.9-12/2001-U.3 of Government of India

Re-accredited by NAAC with 'A++' grade

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1. Title

Working with JSON: Part 1 – JSON using jQuery and JavaScript (Restaurant Orders, Student Details, Employee Management); Part 2 – JSON using fetch() (Student Details and Student Management System).

2. Software / Tools Used

• HTML5, CSS3 and JavaScript (ES6)

• jQuery 3.7.1 (CDN) – $.getJSON(), $.each()

• fetch() API, JSON.stringify(), JSON.parse(), localStorage

• Visual Studio Code, Web Browser (Safari), local web server

3. Program Code

PART 1: JSON

1(a) Restaurant Online Food Orders – jQuery $.getJSON()

orders/index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Restaurant Order Management - Practical 10</title>
    <!-- Simple External CSS -->
    <link rel="stylesheet" href="style.css">
    <!-- 1. Use jQuery library through CDN -->
    <script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>
</head>
<body>

    <div class="container">
        <h2>Restaurant Online Food Orders</h2>
        <p class="desc">Case Study: Dynamic Order Loading using jQuery &amp; JSON</p>

        <!-- View Orders Action Button -->
        <div class="btn-container">
            <button id="viewOrdersBtn">View Orders</button>
        </div>

        <!-- 8. Error message display area -->
        <div id="errorMessage" class="error-message"></div>

        <!-- 3. Dynamic Orders HTML Table -->
        <table id="ordersTable">
            <thead>
                <tr>
                    <th>Order ID</th>
                    <th>Customer Name</th>
                    <th>Food Item</th>
                    <th>Quantity</th>
                    <th>Total Amount</th>
                    <th>Order Status</th>
                </tr>
            </thead>
            <tbody id="ordersTableBody">
                <tr>
```

```
                    <td colspan="6" style="color: #777;">Click "View Orders" above to load all
orders dynamically.</td>
                </tr>
            </tbody>
        </table>

        <!-- 6 & 7. Summary of Total Orders and Total Sales Amount -->
        <div class="summary-box">
            <div>Total Orders: <span id="totalOrdersCount">0</span></div>
            <div>Total Sales Amount: <span id="totalSalesAmount">₹0</span></div>
        </div>
    </div>

    <!-- Main Logic Script -->
    <script src="script.js"></script>
</body>
</html>
```

orders/style.css

```
body {
    font-family: Arial, sans-serif;
    background-color: #f4f6f9;
    margin: 0;
    padding: 30px;
    color: #333;
}

.container {
    max-width: 900px;
    margin: 0 auto;
    background: #ffffff;
    padding: 25px 30px;
    border-radius: 8px;
    box-shadow: 0 2px 8px rgba(0, 0, 0, 0.1);
}

h2 {
    margin-top: 0;
    color: #2c3e50;
    text-align: center;
}

p.desc {
    text-align: center;
    color: #666;
    margin-bottom: 20px;
}

.btn-container {
    text-align: center;
    margin-bottom: 20px;
}

button {
    background-color: #007bff;
    color: white;
    border: none;
    padding: 10px 24px;
    font-size: 16px;
    border-radius: 5px;
    cursor: pointer;
    transition: background-color 0.2s;
}

button:hover {
    background-color: #0056b3;
}

/* Alert / Error Message */
.error-message {
    background-color: #f8d7da;
    color: #721c24;
    border: 1px solid #f5c6cb;
    padding: 12px;
    border-radius: 4px;
    margin-bottom: 15px;
    text-align: center;
    display: none;
}
```

```
/* Table Styling */
table {
    width: 100%;
    border-collapse: collapse;
    margin-top: 10px;
}

th, td {
    border: 1px solid #ddd;
    padding: 12px;
    text-align: center;
}

th {
    background-color: #007bff;
    color: white;
}

tbody tr:nth-child(even) {
    background-color: #f9f9f9;
}

tbody tr:hover {
    background-color: #f1f1f1;
}

/* Status Badges */
.status {
    padding: 4px 8px;
    border-radius: 4px;
    font-weight: bold;
    font-size: 13px;
}

.status-preparing {
    background-color: #fff3cd;
    color: #856404;
}

.status-ready {
    background-color: #d1ecf1;
    color: #0c5460;
}

.status-delivered {
    background-color: #d4edda;
    color: #155724;
}

/* Summary Box */
.summary-box {
    margin-top: 20px;
    padding: 15px 20px;
    background-color: #e9ecef;
    border-radius: 6px;
    display: flex;
    justify-content: space-around;
    font-size: 16px;
    font-weight: bold;
}

.summary-box span {
    color: #007bff;
}
```

orders/script.js

```
$(document).ready(function () {
    $("#viewOrdersBtn").click(function () {
        // Hide previous error message
        $("#errorMessage").hide().text("");

        // 2. Use $.getJSON() to retrieve data from orders.json
        $.getJSON("orders.json", function (data) {
            // 5. Do not allow duplicate rows if the View Orders button is clicked multiple times
            $("#ordersTableBody").empty();

            let totalOrders = 0;
            let totalSales = 0;
```

```
            // 4. Use $.each() to process all the JSON records
            $.each(data, function (index, order) {
                totalOrders++;
                totalSales += Number(order.totalAmount);

                // Set status badge style
                let statusClass = "status";
                let statusLower = (order.orderStatus || "").toLowerCase();
                if (statusLower === "preparing") {
                    statusClass += " status-preparing";
                } else if (statusLower === "ready") {
                    statusClass += " status-ready";
                } else if (statusLower === "delivered") {
                    statusClass += " status-delivered";
                }

                // 3. Display the orders dynamically in an HTML table
                let row = `
                    <tr>
                        <td>${order.orderId}</td>
                        <td>${order.customerName}</td>
                        <td>${order.foodItem}</td>
                        <td>${order.quantity}</td>
                        <td>₹${order.totalAmount}</td>
                        <td><span class="${statusClass}">${order.orderStatus}</span></td>
                    </tr>
                `;
                $("#ordersTableBody").append(row);
            });

            // 6. Display the total number of orders below the table
            $("#totalOrdersCount").text(totalOrders);

            // 7. Calculate and display the total sales amount of all orders
            $("#totalSalesAmount").text("₹" + totalSales);

        }).fail(function (jqXHR, textStatus, errorThrown) {
            // 8. Display an appropriate message if orders.json cannot be loaded
            let msg = "Error: Unable to load 'orders.json'. Please ensure the file is present and
accessible.";
            $("#errorMessage").text(msg).show();
        });
    });
});
```

orders/orders.json

```
[
  { "orderId": "ORD101", "customerName": "Aarav Sharma", "foodItem": "Paneer Butter Masala & Naan",
"quantity": 2, "totalAmount": 480, "orderStatus": "Delivered" },
  { "orderId": "ORD102", "customerName": "Priya Patel", "foodItem": "Veg Hakka Noodles", "quantity":
1, "totalAmount": 320, "orderStatus": "Ready" },
  { "orderId": "ORD103", "customerName": "Rohan Gupta", "foodItem": "Biryani", "quantity": 2,
"totalAmount": 650, "orderStatus": "Preparing" },
  { "orderId": "ORD104", "customerName": "Sneha Reddy", "foodItem": "Margherita Pizza", "quantity":
1, "totalAmount": 399, "orderStatus": "Delivered" },
  { "orderId": "ORD105", "customerName": "Vikram Singh", "foodItem": "Masala Dosa & Coffee",
"quantity": 3, "totalAmount": 360, "orderStatus": "Preparing" }
]
```

1(b) Employee Management – JavaScript & JSON

emp/index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Employee Management - Practical 10b</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <div class="header">
    <h1>Employee Management System</h1>
    <p>Practical 10b: Manage Employee Information using JavaScript &amp;amp; JSON</p>
  </div>

  <div class="layout">
```

```
    <!-- Left: form + JSON output -->
    <div class="card">
      <h2>Add New Employee</h2>
      <form id="empForm">
        <label for="empId">Employee ID:</label>
        <input type="text" id="empId" placeholder="e.g., EMP105">

        <label for="empName">Employee Name:</label>
        <input type="text" id="empName" placeholder="e.g., John Doe">

        <label for="empDept">Department:</label>
        <select id="empDept">
          <option value="">-- Select Department --</option>
          <option>Engineering</option>
          <option>Human Resources</option>
          <option>Finance</option>
          <option>Marketing</option>
        </select>

        <label for="empSalary">Salary (₹ / $):</label>
        <input type="number" id="empSalary" placeholder="e.g., 65000">
        <div id="error" class="err"></div>

        <button type="submit" class="btn-add">Add Employee (Save to JSON)</button>
        <button type="button" id="loadBtn" class="btn-load">Load Employees (fetch JSON)</button>
        <button type="button" id="clearBtn" class="btn-clear">Clear Table View</button>
      </form>

      <div class="json-box">
        <h3>Converted JSON (JSON.stringify)</h3>
        <pre id="jsonOutput">// Click "Load Employees" or add an employee.</pre>
      </div>
    </div>

    <!-- Right: records table -->
    <div class="card">
      <div class="table-head">
        <h2 style="border:0;margin:0;padding:0">Employee Records Table</h2>
        <span class="badge-total" id="totalBadge">Total: 0 Employees</span>
      </div>
      <table>
        <thead>
          <tr><th>#</th><th>Emp ID</th><th>Name</th><th>Department</th><th>Salary</th></tr>
        </thead>
        <tbody id="empBody"></tbody>
        <tfoot>
          <tr><td colspan="4">Total Salary:</td><td style="text-align:left"
id="totalSalary">₹0</td></tr>
        </tfoot>
      </table>
    </div>
  </div>

  <script src="script.js"></script>
</body>
</html>
```

emp/style.css

```
* { box-sizing: border-box; }
body { font-family: "Segoe UI", Arial, sans-serif; background: #f4f6fa; color: #2d3748; margin: 0;
padding: 0 0 30px; }
.header { max-width: 1400px; margin: 0 auto 28px; background: #fff; padding: 24px; text-align:
center; border-radius: 0 0 10px 10px; box-shadow: 0 1px 4px rgba(0,0,0,.08); }
.header h1 { margin: 0 0 8px; font-size: 26px; }
.header p { margin: 0; color: #4a5568; }
.layout { max-width: 1400px; margin: 0 auto; display: grid; grid-template-columns: 430px 1fr; gap:
28px; align-items: start; padding: 0 0; }
.card { background: #fff; border-radius: 10px; padding: 28px; box-shadow: 0 1px 4px rgba(0,0,0,.08);
}
.card h2 { margin: 0 0 20px; font-size: 22px; border-bottom: 2px solid #edf2f7; padding-bottom:
14px; }
label { display: block; font-weight: 600; margin: 14px 0 8px; }
input, select { width: 100%; padding: 12px; border: 1px solid #cbd5e0; border-radius: 6px; font-
size: 16px; }
button { width: 100%; padding: 14px; margin-top: 10px; border: 0; border-radius: 6px; color: #fff;
font-size: 16px; font-weight: 600; cursor: pointer; }
.btn-add { background: #3b69b0; } .btn-load { background: #3f7d58; } .btn-clear { background:
#718096; }
button:hover { opacity: .9; }
```

```
.json-box { background: #1a202c; color: #e2e8f0; border-radius: 8px; padding: 18px; margin-top:
22px; }
.json-box h3 { margin: 0 0 10px; font-size: 15px; color: #a0aec0; text-transform: uppercase; }
.json-box pre { margin: 0; white-space: pre-wrap; font-family: Consolas, monospace; font-size: 14px;
max-height: 260px; overflow: auto; }
.table-head { display: flex; justify-content: space-between; align-items: center; }
.badge-total { background: #ebf4ff; color: #2b6cb0; border: 1px solid #bee3f8; border-radius: 20px;
padding: 6px 14px; font-weight: 600; font-size: 15px; }
table { width: 100%; border-collapse: collapse; }
th { background: #f7fafc; text-align: left; padding: 14px 10px; }
td { padding: 14px 10px; border-bottom: 1px solid #edf2f7; }
.dept { background: #edf2f7; border-radius: 4px; padding: 5px 10px; font-size: 15px; }
tfoot td { background: #f7fafc; font-weight: 700; text-align: right; }
.err { color: #e53e3e; font-size: 14px; margin-top: 6px; }
@media (max-width: 900px) { .layout { grid-template-columns: 1fr; } }
```

emp/script.js

```
// Practical 10b: Manage Employee Information using JavaScript & JSON
const STORAGE_KEY = "employeeRecords";
let employees = [];

const form = document.getElementById("empForm");
const body = document.getElementById("empBody");

// Escape text before inserting it into HTML
function esc(s) {
  return String(s).replace(/[&<>"']/g, c => ({ "&": "&amp;", "<": "&lt;", ">": "&gt;", '"':
"&quot;", "'": "&#39;" }[c]));
}

// Draw table, total badge, total salary and the JSON.stringify output
function render(note) {
  body.innerHTML = "";
  let total = 0;
  employees.forEach((e, i) => {
    total += Number(e.salary);
    body.innerHTML += `
      <tr>
        <td>${i + 1}</td>
        <td><b>${esc(e.id)}</b></td>
        <td>${esc(e.name)}</td>
        <td><span class="dept">${esc(e.department)}</span></td>
        <td>₹${Number(e.salary).toLocaleString("en-IN")}</td>
      </tr>`;
  });
  document.getElementById("totalBadge").textContent = `Total: ${employees.length} Employees`;
  document.getElementById("totalSalary").textContent = "₹" + total.toLocaleString("en-IN");

  // JavaScript object -> JSON string
  document.getElementById("jsonOutput").textContent =
    (note ? note + "\n" : "") + JSON.stringify(employees, null, 2);
}

// Save the array in localStorage as a JSON string
function save() {
  localStorage.setItem(STORAGE_KEY, JSON.stringify(employees));
}

// Add Employee (Save to JSON)
form.addEventListener("submit", function (ev) {
  ev.preventDefault();
  const id = document.getElementById("empId").value.trim();
  const name = document.getElementById("empName").value.trim();
  const department = document.getElementById("empDept").value;
  const salary = parseFloat(document.getElementById("empSalary").value);
  const error = document.getElementById("error");

  if (!id || !name || !department || isNaN(salary) || salary < 0) {
    error.textContent = "Please fill all fields with valid values.";
    return;
  }
  if (employees.some(e => e.id.toLowerCase() === id.toLowerCase())) {
    error.textContent = "Employee ID already exists.";
    return;
  }
  error.textContent = "";
  employees.push({ id, name, department, salary });
  save();
  render("// New employee added and saved:");
```

```
  form.reset();
});

// Load Employees (fetch JSON) - read employees.json, then keep locally added ones
document.getElementById("loadBtn").addEventListener("click", function () {
  fetch("employees.json")
    .then(res => {
      if (!res.ok) throw new Error("HTTP " + res.status);
      return res.json();
    })
    .then(fileData => {
      const saved = JSON.parse(localStorage.getItem(STORAGE_KEY) || "[]");   // JSON string ->
object
      const extra = saved.filter(s => !fileData.some(f => f.id === s.id));
      employees = fileData.concat(extra);
      save();
      render(`// Data loaded from employees.json (${employees.length} records):`);
    })
    .catch(err => {
      document.getElementById("error").textContent =
        "Could not load employees.json. Run the page through a local server.";
      console.error(err);
    });
});

// Clear Table View
document.getElementById("clearBtn").addEventListener("click", function () {
  employees = [];
  localStorage.removeItem(STORAGE_KEY);
  render("// Table cleared.");
});

// Restore saved data on page load
employees = JSON.parse(localStorage.getItem(STORAGE_KEY) || "[]");
render("");
```

emp/employees.json

```
[
  { "id": "EMP101", "name": "Aarav Sharma", "department": "Engineering", "salary": 75000 },
  { "id": "EMP102", "name": "Priya Patel", "department": "Human Resources", "salary": 62000 },
  { "id": "EMP103", "name": "Rohan Mehta", "department": "Finance", "salary": 68000 },
  { "id": "EMP104", "name": "Sneha Gupta", "department": "Marketing", "salary": 58000 },
  { "id": "EMP105", "name": "Kavita Singhania", "department": "Marketing", "salary": 85000 }
]
```

1(c) Student Details – JSON using jQuery

studentjq/index.html

```
<!DOCTYPE html>
<html>
    <head>
        <title>JSON using jQuery</title>

        <script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>

        <style>
            table {
                border-collapse: collapse;
                width: 70%;
                margin-top: 20px;
            }

            th, td {
                border: 1px solid black;
                padding: 10px;
                text-align: center;
            }

            th {
                background-color: lightgray;
            }

            button {
                padding: 10px 20px;
                font-size: 16px;
            }
        </style>
    </head>
```

```
    <body>
        <h2>Student Details</h2>
        <button id="loadBtn">Load Students</button>

        <table id="studentTable">
            <thead>
                <tr>
                    <th>ID</th>
                    <th>Name</th>
                    <th>Department</th>
                    <th>Marks</th>
                </tr>
            </thead>

            <tbody id="tableBody"></tbody>
        </table>

        <script>
            $("#loadBtn").click(function() {
                $.getJSON("students.json", function(data) {
                    $("#tableBody").empty();
                    $.each(data, function(index, student) {
                        $("#tableBody").append(`
                        <tr>
                            <td>${student.id}</td>
                            <td>${student.name}</td>
                            <td>${student.department}</td>
                            <td>${student.marks}</td>
                        </tr>
                            `);
                    }); ;
                }).fail(function() {
                    alert("Error loading JSON data");
                });
            });
        </script>
    </body>
</html>
```

studentjq/students.json

```
[
  {
    "id": 1,
    "name": "Rahul",
    "department": "CSE",
    "marks": 85
  },
  {
    "id": 2,
    "name": "Priya",
    "department": "AIML",
    "marks": 92
  },
  {
    "id": 3,
    "name": "Amit",
    "department": "Data Science",
    "marks": 88
  }
]
```

PART 2: fetch()

2(a) Student Details using fetch()

studentfetch/index.html

```
<!DOCTYPE html>
<html>
    <head>
        <title>Student Details</title>

        <style>
            table {
                border-collapse: collapse;
                width: 70%;
                margin-top: 20px;
            }
```

```
            th, td {
                border: 1px solid black;
                padding: 10px;
            }

            th {
                background-color: lightgray;
                text-align: left;
            }
        </style>
    </head>

    <body>
        <h2>Student Details using fetch()</h2>
        <button id="loadBtn">Load Students</button>

        <table id="studentTable">
            <thead>
                <tr>
                    <th>ID</th>
                    <th>Name</th>
                    <th>Department</th>
                    <th>Marks</th>
                </tr>
            </thead>
            <tbody id="tableBody"></tbody>
        </table>

        <script>
            document.getElementById("loadBtn").addEventListener("click", function () {
                // Use fetch() to read students.json
                fetch("students.json")
                    .then(function (response) {
                        if (!response.ok) {
                            throw new Error("HTTP error " + response.status);
                        }
                        return response.json();
                    })
                    .then(function (data) {
                        const tbody = document.getElementById("tableBody");
                        tbody.innerHTML = "";   // avoid duplicate rows on repeated clicks

                        data.forEach(function (student) {
                            tbody.innerHTML += `
                                <tr>
                                    <td>${student.id}</td>
                                    <td>${student.name}</td>
                                    <td>${student.department}</td>
                                    <td>${student.marks}</td>
                                </tr>`;
                        });
                    })
                    .catch(function (error) {
                        console.error(error);
                        alert("Error loading JSON data");
                    });
            });
        </script>
    </body>
</html>
```

studentfetch/students.json

```
[
  {
    "id": 1,
    "name": "Rahul",
    "department": "CSE",
    "marks": 85
  },
  {
    "id": 2,
    "name": "Priya",
    "department": "AIML",
    "marks": 92
  },
  {
    "id": 3,
    "name": "Amit",
```

```
    "department": "Data Science",
    "marks": 88
  }
]
```

2(b) Student Management System (fetch + localStorage)

sms/index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Student Management System</title>
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link href="https://fonts.googleapis.com/css2?
family=Plus+Jakarta+Sans:wght@400;500;600;700;800&display=swap" rel="stylesheet">

    <style>
        * {
            box-sizing: border-box;
            margin: 0;
            padding: 0;
            font-family: 'Plus Jakarta Sans', -apple-system, BlinkMacSystemFont, 'Segoe UI', Roboto,
sans-serif;
        }

        body {
            background-color: #f0f4f8;
            min-height: 100vh;
            display: flex;
            justify-content: center;
            align-items: flex-start;
            padding: 50px 20px;
        }

        .main-container {
            width: 100%;
            max-width: 900px;
            background: #ffffff;
            border-radius: 16px;
            box-shadow: 0 10px 30px rgba(0, 0, 0, 0.05);
            padding: 36px 40px;
        }

        .title {
            text-align: center;
            font-size: 1.85rem;
            font-weight: 800;
            color: #1e293b;
            margin-bottom: 28px;
            display: flex;
            align-items: center;
            justify-content: center;
            gap: 10px;
        }

        /* Form Box */
        .form-box {
            border: 1px solid #e2e8f0;
            border-radius: 12px;
            padding: 24px;
            margin-bottom: 30px;
            background: #ffffff;
        }

        .inputs-row {
            display: grid;
            grid-template-columns: repeat(4, 1fr);
            gap: 16px;
            margin-bottom: 20px;
        }

        .input-group {
            display: flex;
            flex-direction: column;
            gap: 8px;
```

```
        }

        .input-group label {
            font-size: 0.88rem;
            font-weight: 700;
            color: #334155;
        }

        .input-group input {
            width: 100%;
            padding: 10px 14px;
            border: 1px solid #cbd5e1;
            border-radius: 8px;
            font-size: 0.95rem;
            color: #1e293b;
            outline: none;
            transition: all 0.2s ease;
            background: #ffffff;
        }

        .input-group input:focus {
            border-color: #2563eb;
            box-shadow: 0 0 0 3px rgba(37, 99, 235, 0.15);
        }

        .input-group input::placeholder {
            color: #94a3b8;
        }

        /* Button Row */
        .buttons-row {
            display: flex;
            align-items: center;
            gap: 14px;
            flex-wrap: wrap;
        }

        .btn {
            display: inline-flex;
            align-items: center;
            justify-content: center;
            gap: 8px;
            padding: 11px 22px;
            font-size: 0.95rem;
            font-weight: 700;
            border-radius: 8px;
            border: none;
            cursor: pointer;
            transition: all 0.2s ease;
        }

        .btn-add {
            background-color: #2563eb;
            color: #ffffff;
        }

        .btn-add:hover {
            background-color: #1d4ed8;
            box-shadow: 0 4px 12px rgba(37, 99, 235, 0.25);
        }

        .btn-load {
            background-color: #475569;
            color: #ffffff;
        }

        .btn-load:hover {
            background-color: #334155;
            box-shadow: 0 4px 12px rgba(71, 85, 105, 0.25);
        }

        .btn-clear {
            background-color: #ef4444;
            color: #ffffff;
        }

        .btn-clear:hover {
            background-color: #dc2626;
            box-shadow: 0 4px 12px rgba(239, 68, 68, 0.25);
```

```
}

/* Table */
.table-responsive {
    width: 100%;
    overflow-x: auto;
}

table {
    width: 100%;
    border-collapse: collapse;
    text-align: left;
}

thead tr {
    background-color: #f1f5f9;
}

th {
    padding: 14px 20px;
    font-size: 0.82rem;
    font-weight: 800;
    color: #334155;
    text-transform: uppercase;
    letter-spacing: 0.6px;
}

td {
    padding: 18px 20px;
    font-size: 0.95rem;
    color: #1e293b;
    border-bottom: 1px solid #f1f5f9;
    vertical-align: middle;
}

tbody tr:last-child td {
    border-bottom: none;
}

.id-cell {
    font-weight: 800;
    color: #1e293b;
}

.btn-delete {
    background-color: #fee2e2;
    color: #ef4444;
    border: none;
    border-radius: 6px;
    padding: 6px 14px;
    font-size: 0.85rem;
    font-weight: 700;
    cursor: pointer;
    transition: all 0.15s ease;
}

.btn-delete:hover {
    background-color: #ef4444;
    color: #ffffff;
}

.empty-row {
    text-align: center;
    color: #64748b;
    padding: 30px;
    font-size: 0.95rem;
}

@media (max-width: 768px) {
    .inputs-row {
        grid-template-columns: repeat(2, 1fr);
    }
}

@media (max-width: 520px) {
    .inputs-row {
        grid-template-columns: 1fr;
    }
    .buttons-row .btn {
```

```
                width: 100%;
            }
            .main-container {
                padding: 24px 16px;
            }
        }
    </style>
</head>
<body>

    <main class="main-container">
        <!-- Header -->
        <h1 class="title">🎓 Student Management System</h1>

        <!-- Form Card -->
        <section class="form-box">
            <form id="studentForm" onsubmit="handleAddStudent(event)">
                <div class="inputs-row">
                    <div class="input-group">
                        <label for="studentId">Student ID</label>
                        <input type="number" id="studentId" placeholder="e.g. 101" required min="1">
                    </div>
                    <div class="input-group">
                        <label for="studentName">Name</label>
                        <input type="text" id="studentName" placeholder="e.g. Rahul Sharma"
required>
                    </div>
                    <div class="input-group">
                        <label for="studentDept">Department</label>
                        <input type="text" id="studentDept" placeholder="e.g. CSE" required>
                    </div>
                    <div class="input-group">
                        <label for="studentMarks">Marks</label>
                        <input type="number" id="studentMarks" placeholder="e.g. 85" required
min="0" max="100">
                    </div>
                </div>

                <div class="buttons-row">
                    <button type="submit" class="btn btn-add">
                        <span>+</span> Add Student
                    </button>
                    <button type="button" class="btn btn-load" onclick="loadFromStudentsJSON()">
                        📁 Load from students.json
                    </button>
                    <button type="button" class="btn btn-clear" onclick="clearAllData()">
                        🗑️ Clear All
                    </button>
                </div>
            </form>
        </section>

        <!-- Students Table -->
        <div class="table-responsive">
            <table>
                <thead>
                    <tr>
                        <th>ID</th>
                        <th>NAME</th>
                        <th>DEPARTMENT</th>
                        <th>MARKS</th>
                        <th>ACTION</th>
                    </tr>
                </thead>
                <tbody id="studentTableBody">
                    <!-- Dynamic Rows Will Appear Here -->
                </tbody>
            </table>
        </div>
    </main>

    <script>
        const STORAGE_KEY = "studentRecords";

        // Load data on page load
        document.addEventListener("DOMContentLoaded", () => {
            const saved = localStorage.getItem(STORAGE_KEY);
            if (saved) {
                renderTable(JSON.parse(saved));
```

```
        } else {
            // Auto-load default records from students.json on first visit
            loadFromStudentsJSON(false);
        }
    });

    /**
     * Add a new student and store in LocalStorage
     */
    function handleAddStudent(event) {
        event.preventDefault();

        const id = parseInt(document.getElementById("studentId").value.trim());
        const name = document.getElementById("studentName").value.trim();
        const department = document.getElementById("studentDept").value.trim();
        const marks = parseInt(document.getElementById("studentMarks").value.trim());

        if (!id || !name || !department || isNaN(marks)) {
            alert("Please fill in all fields.");
            return;
        }

        let students = getStoredStudents();

        // Check if student with same ID exists
        const existingIndex = students.findIndex(s => Number(s.id) === id);
        if (existingIndex !== -1) {
            students[existingIndex] = { id, name, department, marks };
        } else {
            students.push({ id, name, department, marks });
        }

        // Save to LocalStorage
        saveStudents(students);

        // Re-render table
        renderTable(students);

        // Reset form
        document.getElementById("studentForm").reset();
        document.getElementById("studentId").focus();
    }

    /**
     * Load data from students.json file
     */
    function loadFromStudentsJSON(showAlert = true) {
        fetch("students.json")
            .then(res => {
                if (!res.ok) throw new Error("HTTP " + res.status);
                return res.json();
            })
            .then(data => {
                saveStudents(data);
                renderTable(data);
            })
            .catch(err => {
                console.error("Error loading students.json:", err);
                if (showAlert) {
                    alert("Could not load students.json. Make sure the server is running.");
                }
            });
    }

    /**
     * Delete a single student
     */
    function deleteStudent(id) {
        let students = getStoredStudents();
        students = students.filter(s => Number(s.id) !== Number(id));
        saveStudents(students);
        renderTable(students);
    }

    /**
     * Clear all stored data
     */
    function clearAllData() {
        if (confirm("Are you sure you want to clear all student records?")) {
```

```
                localStorage.removeItem(STORAGE_KEY);
                renderTable([]);
            }
        }

        /**
         * Helper to get data from LocalStorage
         */
        function getStoredStudents() {
            const data = localStorage.getItem(STORAGE_KEY);
            return data ? JSON.parse(data) : [];
        }

        /**
         * Helper to save data to LocalStorage
         */
        function saveStudents(data) {
            localStorage.setItem(STORAGE_KEY, JSON.stringify(data));
        }

        /**
         * Render table rows
         */
        function renderTable(students) {
            const tbody = document.getElementById("studentTableBody");
            tbody.innerHTML = "";

            if (!students || students.length === 0) {
                tbody.innerHTML = `
                    <tr>
                        <td colspan="5" class="empty-row">No student records found. Add a student
above or load from students.json.</td>
                    </tr>
                `;
                return;
            }

            students.forEach(student => {
                const tr = document.createElement("tr");
                tr.innerHTML = `
                    <td class="id-cell">#${student.id}</td>
                    <td>${escapeHTML(student.name)}</td>
                    <td>${escapeHTML(student.department)}</td>
                    <td>${student.marks}</td>
                    <td>
                        <button class="btn-delete"
onclick="deleteStudent(${student.id})">Delete</button>
                    </td>
                `;
                tbody.appendChild(tr);
            });
        }

        function escapeHTML(str) {
            return String(str)
                .replace(/&/g, "&amp;")
                .replace(/</g, "&lt;")
                .replace(/>/g, "&gt;")
                .replace(/"/g, "&quot;")
                .replace(/'/g, "&#039;");
        }
    </script>
</body>
</html>
```

sms/students.json

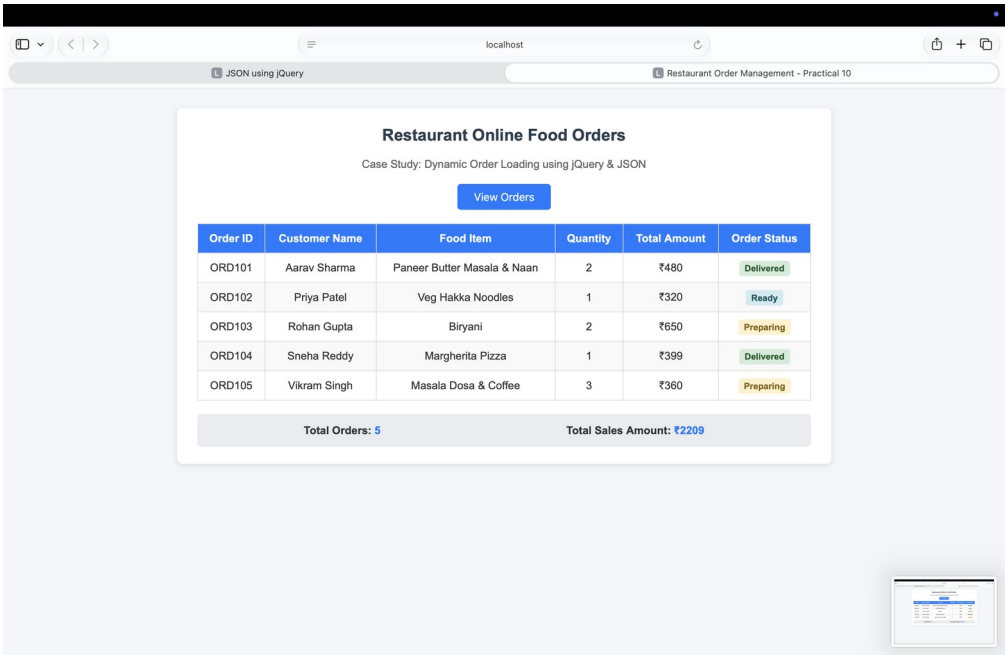
```
[
  {
    "id": 1,
    "name": "Rahul",
    "department": "CSE",
    "marks": 85
  },
  {
    "id": 2,
    "name": "Priya",
    "department": "AIML",
    "marks": 92
  },
```

```
  {
    "id": 3,
    "name": "Amit",
    "department": "Data Science",
    "marks": 88
  }
]
```

4. Output

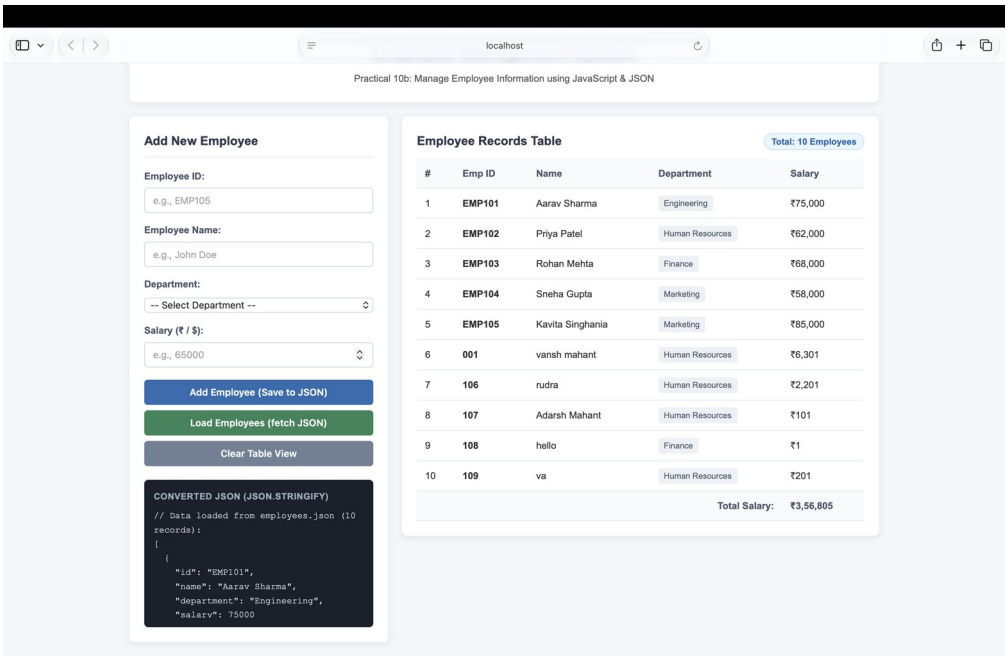
PART 1: JSON

1(a) Restaurant Online Food Orders – 5 orders loaded, Total Orders: 5, Total Sales Amount: ₹2209.



PRN: 24070521038 | Name: Vansh Mahant | File: orders/index.html

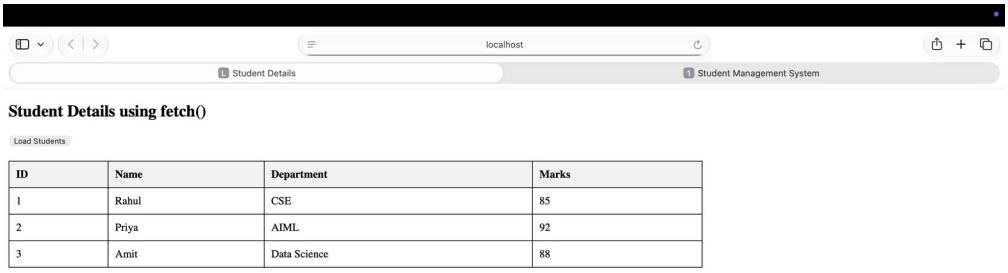
1(b) Employee Management – 10 employees loaded, Total Salary: ₹3,56,805, with the JSON.stringify output.



PRN: 24070521038 | Name: Vansh Mahant | File: emp/index.html

PART 2: fetch()

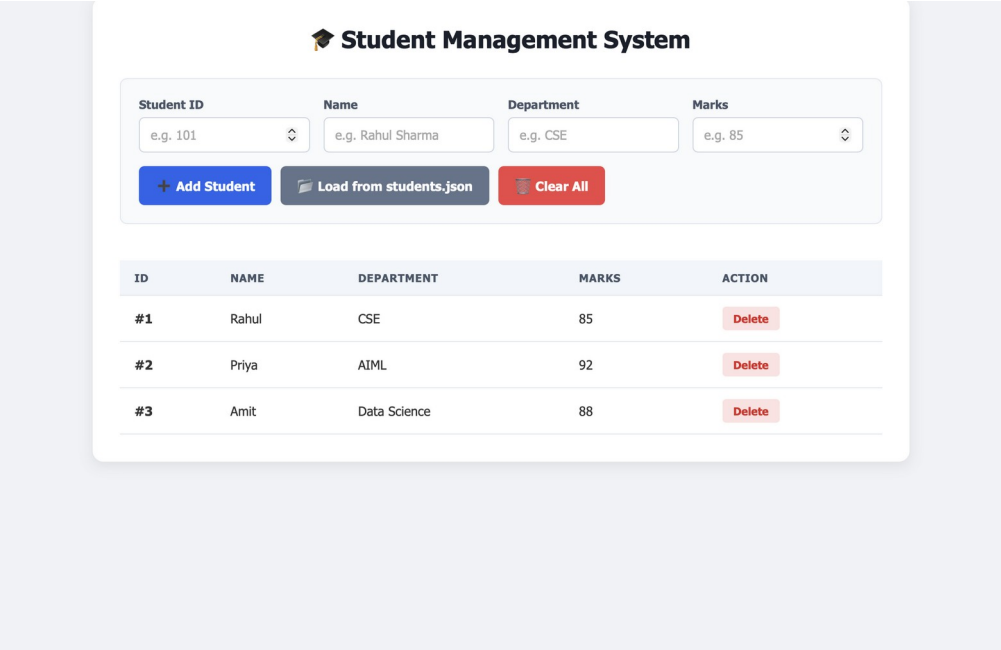
2(a) Student Details using fetch() – data loaded from students.json.



ID	Name	Department	Marks
1	Rahul	CSE	85
2	Priya	AIML	92
3	Amit	Data Science	88

PRN: 24070521038 | Name: Vansh Mahant | File: studentfetch/index.html

2(b) Student Management System – Add, Load from students.json, Clear All and per-row Delete.



PRN: 24070521038 | Name: Vansh Mahant | File: sms/index.html

5. Result / Conclusion

JSON data was loaded and displayed dynamically using both jQuery ($.getJSON with $.each) and the fetch() API with Promises. Data was converted with JSON.stringify() and JSON.parse() and stored in localStorage. Error handling was done with .fail() and .catch(). Totals (order count, sales amount, salary) were calculated and shown correctly.